EXPERIMENT 6: BYTE STUFFING

USING PYTHON

24/7/26

Aim:

To implement byte stuffing using python

byte_stuffing.py - C:/Users/REC/AppData/Local/Programs/Python/Python310/byte_stuffing.p...

File Edit Format Run Options Window Help

```
# Byte Stuffing with Binary Representation

flag = input("Enter Flag Character: ")
escape = input("Enter Escape Character: ")
data = input("Enter Data: ")

stuffed = ""

for ch in data:
    if ch == flag or ch == escape:
        stuffed += escape
    stuffed += ch

frame = flag + stuffed + flag

print("\nOriginal Data      :", data)
print("Stuffed Data       :", stuffed)
print("Transmitted Frame  :", frame)

# Function to print binary values
def show_binary(text):
    for ch in text:
        print(ch, "->", format(ord(ch), '08b'))

print("\nBinary of Flag:")
show_binary(flag)

print("\nBinary of Escape:")
show_binary(escape)

print("\nBinary of Original Data:")
show_binary(data)

print("\nBinary of Stuffed Data:")
show_binary(stuffed)

print("\nBinary of Transmitted Frame:")
show_binary(frame)
```

Ln: 39 Col: 0

IDLE Shell 3.10.6

File Edit Shell Debug Options Window Help

```
>>> = RESTART: C:/Users/REC/AppData/Local/Programs/Python/Python310/byte_stuffing.py
Enter Flag Character: F
Enter Escape Character: E
Enter Data: AEFB

Original Data       : AEFB
Stuffed Data        : AEEEFB
Transmitted Frame   : FAEEEFBF

Binary of Flag:
F -> 01000110

Binary of Escape:
E -> 01000101

Binary of Original Data:
A -> 01000001
E -> 01000101
F -> 01000110
B -> 01000010

Binary of Stuffed Data:
A -> 01000001
E -> 01000101
E -> 01000101
E -> 01000101
F -> 01000110
B -> 01000010

Binary of Transmitted Frame:
F -> 01000110
A -> 01000001
E -> 01000101
E -> 01000101
E -> 01000101
F -> 01000110
B -> 01000010
F -> 01000110
>>>
```

Ln: 42 Col: 0

RESULT:

Thus, implementation of byte stuffing using python is executed successfully.